Logistic regression

We have responses $Y_1, \ldots, Y_n$ independent random variables (conditional on the covariates) where

$$Y_i \sim \text{Bernoulli}(\pi(\underline{x}; \underline{\beta}))$$

($\underline{\beta}$: parameter β)

The function $\pi(\cdot; \theta)$ should map $\mathbb{R}$ (or $\mathbb{R}^d$) to $[0, 1]$.

$$\underbrace{\log\left(\left[\frac{\pi(\underline{x};\theta)}{1-\pi(\underline{x};\theta)}\right]\right)}_{\text{logit}(\pi(x;\theta))} = \underbrace{\beta_0 + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1}}_{\text{linear predictor}}$$

([ ] : odds)

What does this mean for probabilities?

Suppose $\eta = \log\left(\frac{p}{1-p}\right)$ $\quad p \in [0,1]$

$$\Rightarrow \quad \exp(\eta) = \frac{p}{1-p} \quad \Rightarrow \quad \exp(\eta)(1-p) = p$$

$$\Rightarrow \quad p = \frac{\exp(\eta)}{1+\exp(\eta)}$$

As $\eta \to \infty$, $p \to 1$

As $\eta \to -\infty$, $p \to 0$

For any finite η, $p \in (0,1)$

In the logistic model

$$\pi(\underline{x}; \underline{\beta}) = \frac{\exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1})}{1 + \exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1})}.$$

Returning to odds

$$\frac{p}{1-p} = \exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1})$$

$$\text{odds} = \exp(\beta_0) \times \exp(\beta_1 x_1) \cdots \exp(\beta_{p-1} x_{p-1}).$$

A one unit increase in x_1, with all other covariates held constant, changes the odds by a factor of $\exp(\beta_1)$.

Maximum likelihood estimation

Let's introduce functions $F(z) = \dfrac{e^z}{1+e^z}$ and

$$f(z) = F'(z) = \frac{e^z(1+e^z) - e^z \cdot e^z}{(1+e^z)^2} = \frac{e^z}{(1+e^z)^2} = F(z) \cdot (1 - F(z))$$

Note $\dfrac{F(z)}{1-F(z)} = e^z$

The probabilities in the logistic regression are given by $F(\underline{x}\beta)$ (thinking of $\underline{x}$ as a row vector and β as a column vector)

The likelihood function is then

$$\mathcal{L}(\beta) = \prod_{i=1}^{n} \pi(\underline{x}_i;\beta)^{y_i} (1 - \pi(\underline{x}_i;\beta))^{1-y_i} = \prod_{i=1}^{n} F(\underline{x}_i\beta)^{y_i} (1 - F(\underline{x}_i\beta))^{1-y_i}$$

The log likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \left\{ y_i \log(F(\underline{x}_i\beta)) + (1-y_i) \log(1 - F(\underline{x}_i\beta)) \right\}$$

$$= \sum_{i=1}^{n} \left\{ \log(1 - F(\underline{x}_i\beta)) + y_i \log\left(\frac{F(\underline{x}_i\beta)}{1 - F(\underline{x}_i\beta)} \right) \right\}$$

$$= \sum_{i=1}^{n} \left\{ \log(1 - F(\underline{x}_i\beta)) + y_i \times \underline{x}_i\beta \right\}$$

Differentiate $\ell(\underline{\beta})$ to get the score function

$$\frac{\partial \ell(\underline{\beta})}{\partial \beta_j} = \sum_{i=1}^{n} \left\{ \frac{1}{1-F(\underline{x}_i\underline{\beta})} \times (-f(\underline{x}_i\underline{\beta})\, x_{ij} + y_i\, x_{ij} \right\}$$

$$= \sum_{i=1}^{n} \left\{ y_i x_{ij} - x_{ij} F(\underline{x}_i\underline{\beta}) \right\}$$

$$= \sum_{i=1}^{n} x_{ij} \left\{ y_i - F(\underline{x}_i\underline{\beta}) \right\}$$

The MLE is found by solving the nonlinear system of equations

$$\sum_{i=1}^{n} x_{ij} \left\{ y_i - F(\underline{x}_i\underline{\beta}) \right\} = 0 \quad \forall\, j = 0, \ldots p-1.$$

The solution can be found using "Iteratively reweighted least squares"

We want to calculate the observed information matrix

$$\frac{\partial^2 \ell(\underline{\beta})}{\partial \beta_j \partial \beta_k} = \sum_{i=1}^{n} -x_{ij}\, F(\underline{x}_i\underline{\beta})\,(1-F(\underline{x}_i\underline{\beta}))\, x_{ik}$$

$$= -\left(\underline{\underline{X}}^T \underline{\underline{W}}\, \underline{\underline{X}}\right)_{jk}$$

$\underline{\underline{X}}$: $n \times p$ matrix of covariates, row i corresponds to $\underline{x}_i$

where $\underline{\underline{W}}$ is the diagonal matrix with $W_{ii} = F(\underline{x}_i\underline{\beta})(1-F(\underline{x}_i\underline{\beta}))$

So the observed information matrix is

$$J(\underline{\beta}) = \underline{\underline{X}}^T \underline{\underline{W}}\, \underline{\underline{X}}$$

As this does not depend on the response $Y_1, \ldots, Y_n$ the observed and expected information are equal

$$I(\underline{\beta}) = \mathbb{E}_R\, J(\underline{\beta}) = \underline{\underline{X}}^T \underline{\underline{W}}\, \underline{\underline{X}}.$$

Asymptotically

$$\hat{\underline{\beta}} \sim N\left(\underline{\beta}, (\underline{X}^T \underline{W} \underline{X})^{-1}\right)$$

So standard errors for $\hat{\beta}_j$ is obtained by

$$\sqrt{(\underline{X}^T \underline{W} \underline{X})^{-1}_{jj}}$$

replacing the $\underline{\beta}$ in $\underline{W}$ by the MLE $\hat{\underline{\beta}}$.

Predictions

The predicted probability for a new vector of covariates is

$$\frac{\exp(\underline{x}_{new} \hat{\underline{\beta}})}{1 + \exp(\underline{x}_{new} \hat{\underline{\beta}})} = F(\underline{x}_{new} \hat{\underline{\beta}})$$

The standard error of $\underline{x}_{new} \hat{\underline{\beta}}$

$$\text{var}(\underline{x}_{new} \hat{\underline{\beta}}) = \underline{x}_{new} \, \text{cov}(\hat{\underline{\beta}}) \, \underline{x}_{new}^T$$

$$= \underline{x}_{new} (\underline{X}^T \underline{W} \underline{X})^{-1} \underline{x}_{new}^T$$

$$\text{se}(\underline{x}_{new} \hat{\underline{\beta}}) = \sqrt{\underline{x}_{new} (\underline{X}^T \underline{W} \underline{X})^{-1} \underline{x}_{new}^T}$$

For the predicted probability

$$\text{Var}(F(\underline{x}_{new} \hat{\underline{\beta}})) \approx \left[F(\underline{x}_{new} \hat{\underline{\beta}}) (1 - F(\underline{x}_{new} \underline{\beta}))\right]^2 \cdot \text{var}(\underline{x}_{new} \hat{\underline{\beta}})$$

Basic diagnostics

raw residuals $y_i - \hat{p}_i$, where $\hat{p}_i$ is the fitted probability for the i-th observation. Plots ~~the~~ of the raw residual against fitted values / linear predictor / ~~exp~~ individual covariates — are not very informative.

Binned plots of raw residuals can be useful. Points are grouped / binned along horizontal axis and average of raw residual in that bin is plotted. If the logistic model is appropriate these should have mean 0. Grey lines give $\pm 2\,SE$, where SE are determined the standard error of the mean in the bin. $\sim 95\%$ of points should fall within the grey lines.

Leverage is similar to linear regression leverages. The hat matrix becomes

$$\underset{=}{H} = \underset{=}{W}^{1/2}\,\underset{=}{X}\,(\underset{=}{X}^T\underset{=}{W}\underset{=}{X})^{-1}\,\underset{=}{X}^T\,\underset{=}{W}^{1/2}$$

This depends on both covariates and weights

As a guide observations with $H_{ii} > 2p/n$ should be investigated more closely.